

LLM + VLM Autonomous Drone Project

Best Learning Roadmap

Recommended Learning Order

1. Andrej Karpathy – Intro to LLMs
2. 3Blue1Brown – Transformers
3. Carbon Aeronautics – How Quadcopters Work
4. Brian Douglas – PID Controllers
5. OpenCV Full Course (freeCodeCamp)
6. Google RT-2
7. PaLM-E
8. LLaVA / CLIP Explained
9. Bitcraze Crazyflie Tutorials
10. ROS2 by Robotics Back-End

Reference Links

Andrej Karpathy – Intro to LLMs

https://www.youtube.com/results?search_query=Andrej+Karpathy+Intro+to+Large+Language+Models

Build GPT from Scratch

https://www.youtube.com/results?search_query=Andrej+Karpathy+Let's+build+GPT+from+scratch

3Blue1Brown – Transformers

https://www.youtube.com/results?search_query=3Blue1Brown+Transformer

StatQuest – Transformers

https://www.youtube.com/results?search_query=StatQuest+Transformers

Carbon Aeronautics – Quadcopters

https://www.youtube.com/results?search_query=Carbon+Aeronautics+How+Quadcopters+Work

Brian Douglas – PID

https://www.youtube.com/results?search_query=Brian+Douglas+PID

OpenCV Full Course

https://www.youtube.com/results?search_query=OpenCV+Full+Course+freeCodeCamp

YOLO Tutorial

https://www.youtube.com/results?search_query=YOLO+Object+Detection+tutorial

LLaVA

https://www.youtube.com/results?search_query=LLaVA+explained

CLIP

https://www.youtube.com/results?search_query=OpenAI+CLIP+explained

Google RT-2

https://www.youtube.com/results?search_query=Google+RT-2+robotics

PaLM-E

https://www.youtube.com/results?search_query=PaLM-E+explained

NVIDIA GR00T

https://www.youtube.com/results?search_query=NVIDIA+GR00T

ROS2

https://www.youtube.com/results?search_query=Robotics+Back-End+ROS2

Bitcraze Crazyflie

https://www.youtube.com/results?search_query=Bitcraze+Crazyflie+tutorial

Function Calling

https://www.youtube.com/results?search_query=OpenAI+Function+Calling

LangGraph

https://www.youtube.com/results?search_query=LangGraph+tutorial

CrewAI

https://www.youtube.com/results?search_query=CrewAI+tutorial